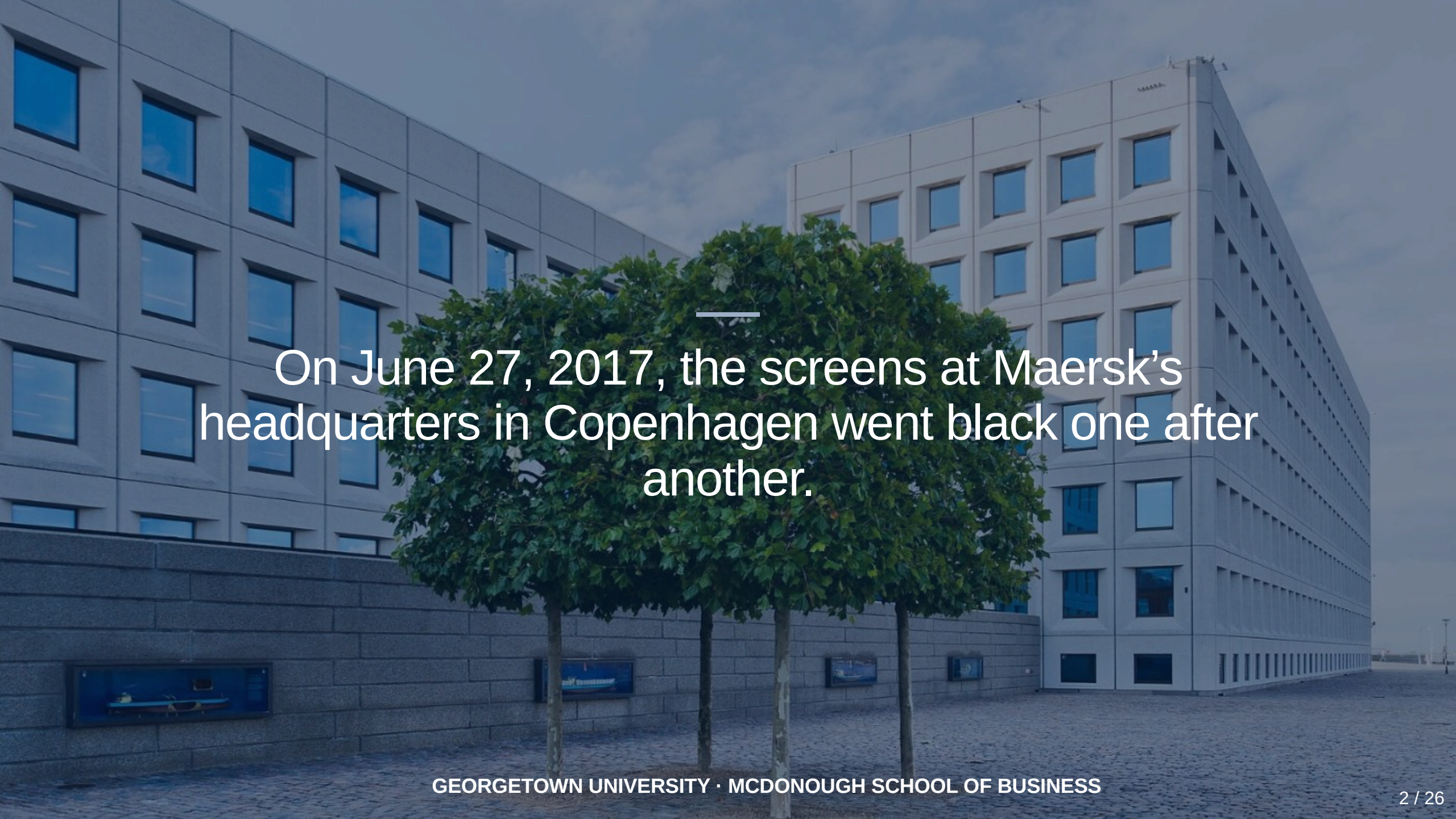
When a Digital Shock Hits a Global Firm

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On June 27, 2017, the screens at Maersk’s headquarters in Copenhagen went black one after another.

YOUR IB VOICE

FROM UKRAINE TO NEW JERSEY

NotPetya was aimed at Ukraine, yet it stopped Maersk’s terminals from New Jersey to Mumbai. Using Greenberg, explain what about the way Maersk was organized let an attack on one country stop a global firm.

10 minutes · Individual · Notes permitted

Why did an attack on Ukraine stop a global shipping company?

Maersk neglected its security and ran old software.

Maersk ran one network worldwide, so an infection anywhere could reach everywhere.

Russia meant to punish any firm that did business in Ukraine.

No firm can defend itself against a government’s military hackers.

> Which of these could Maersk itself have changed?

How did a Ukrainian tax program reach Copenhagen?

NotPetya spread faster than anyone could unplug



Greenberg, “The Untold Story of NotPetya,” *WIRED*, 2018.

Russian military hackers hid NotPetya in an update to M.E.Doc, the tax software most Ukrainian firms use.

It used a tool stolen from the NSA, and passwords lifted from memory, to jump between computers.

> The ransom note was a disguise, because no key existed to unlock the files.

In Ukraine, banks, airports, and ministries went dark



ECAC, Volodymyr Omelyan, July 2018 · Wikimedia Commons, CC0

The attack hit more than 22 Ukrainian banks, two airports, and almost every federal agency.

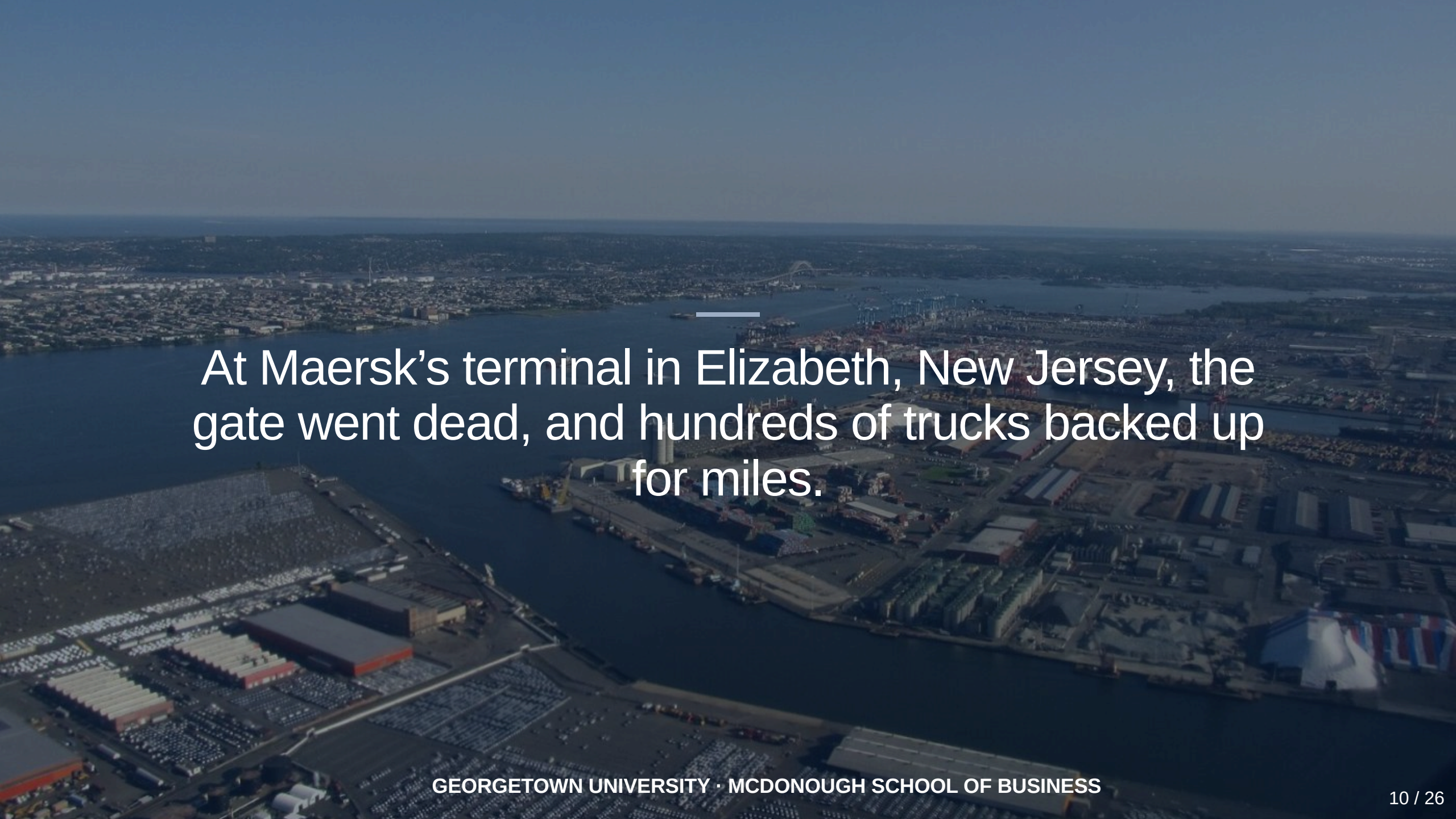
One official estimated that it wiped a tenth of the country's computers.

> "The government was dead," said Volodymyr Omelyan, Ukraine's minister of infrastructure.

Cisco’s Craig Williams argues the damage abroad was intended: “If you do business in Ukraine, bad things are going to happen to you.” Others call it collateral damage. Does the answer change what firms should do?

> Should a firm treat its office in a country at war as a risk to its whole network?

What did one infected computer cost Maersk?



At Maersk’s terminal in Elizabeth, New Jersey, the gate went dead, and hundreds of trucks backed up for miles.

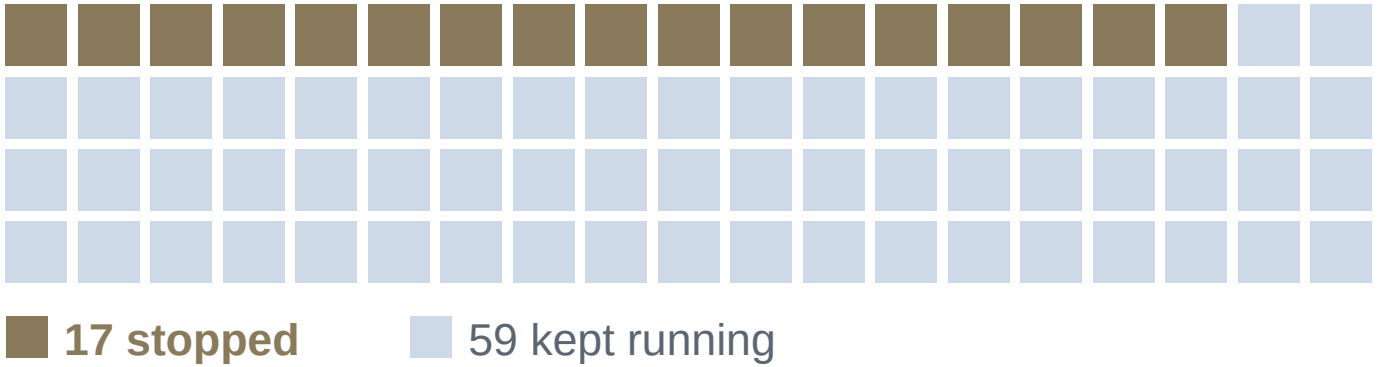
Seventeen of Maersk’s 76 terminals stopped

The worm entered through one computer in Maersk’s office in Odessa that ran M.E.Doc.

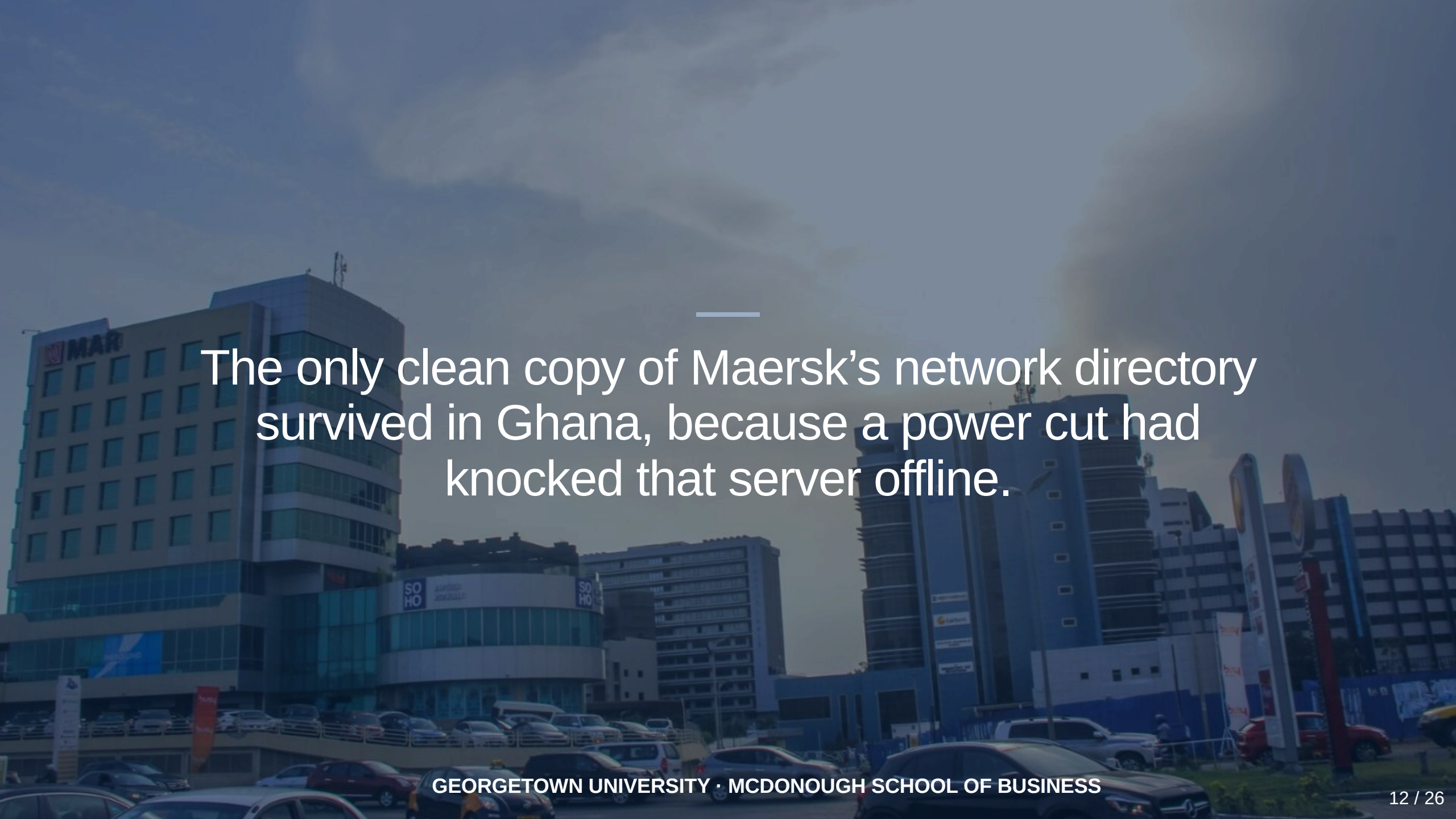
The ships’ computers were fine, but the terminals lost the files listing what each ship carried.

> *Digital interdependence* means a firm’s ships, cranes, and gates stop when the software they share stops.

APM TERMINALS, JUNE 27, 2017



Greenberg, “The Untold Story of NotPetya,” *WIRED*, 2018.



The only clean copy of Maersk’s network directory survived in Ghana, because a power cut had knocked that server offline.

Maersk approved a security fix and never built it



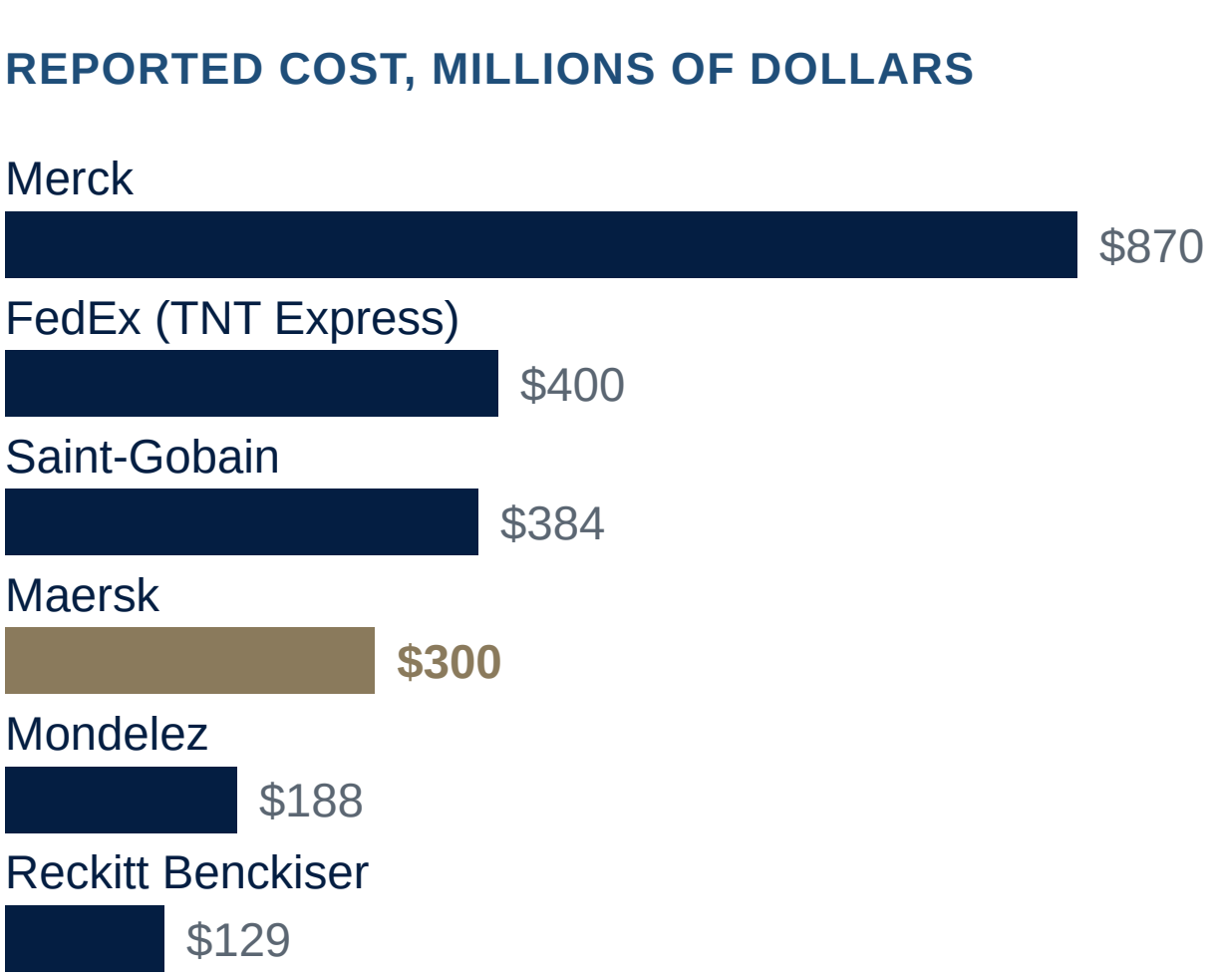
World Economic Forum, Jim Hagemann Snabe, 2013 · Wikimedia Commons, CC BY-SA 2.0

In 2016 Maersk's IT executives warned that its network was not divided into sections.

The fix was approved and budgeted, but no one's bonus depended on it.

> *Operational resilience* is a firm's ability to keep working through a failure and recover quickly.

The attack cost firms far from Ukraine billions



Greenberg, “The Untold Story of NotPetya,” *WIRED*, 2018: damages reported by the companies.

The White House put the total damage above $10 billion.

Many victims without public shareholders never reported their losses.

> *Cyber risk* is the chance that an attack on a firm’s computers stops its business.

Maersk’s IT chiefs were paid for other targets, so the security fix waited. Who in a firm should be paid to prevent a disaster that may never happen?

> How is skipping this fix like keeping no spare stock?

Who pays when a cyberattack is an act of war?

Insurers called NotPetya an act of war

CLAIM	AMOUNT	OUTCOME
Mondelez v. Zurich	about $100 million	Settled, 2022
Merck v. Ace American	about $1.4 billion	**War clause rejected, 2023** Settled, 2024

Court filings and reporting by *The Record* (2022) and *Cybersecurity Dive* (2024); New Jersey Superior Court, Appellate Division, May 1, 2023.

In February 2018 the U.S. government said Russia's military launched the attack. Insurers refused some claims under clauses that exclude war.

> Since 2023 Lloyd's of London has required its cyber policies to exclude attacks backed by a state.

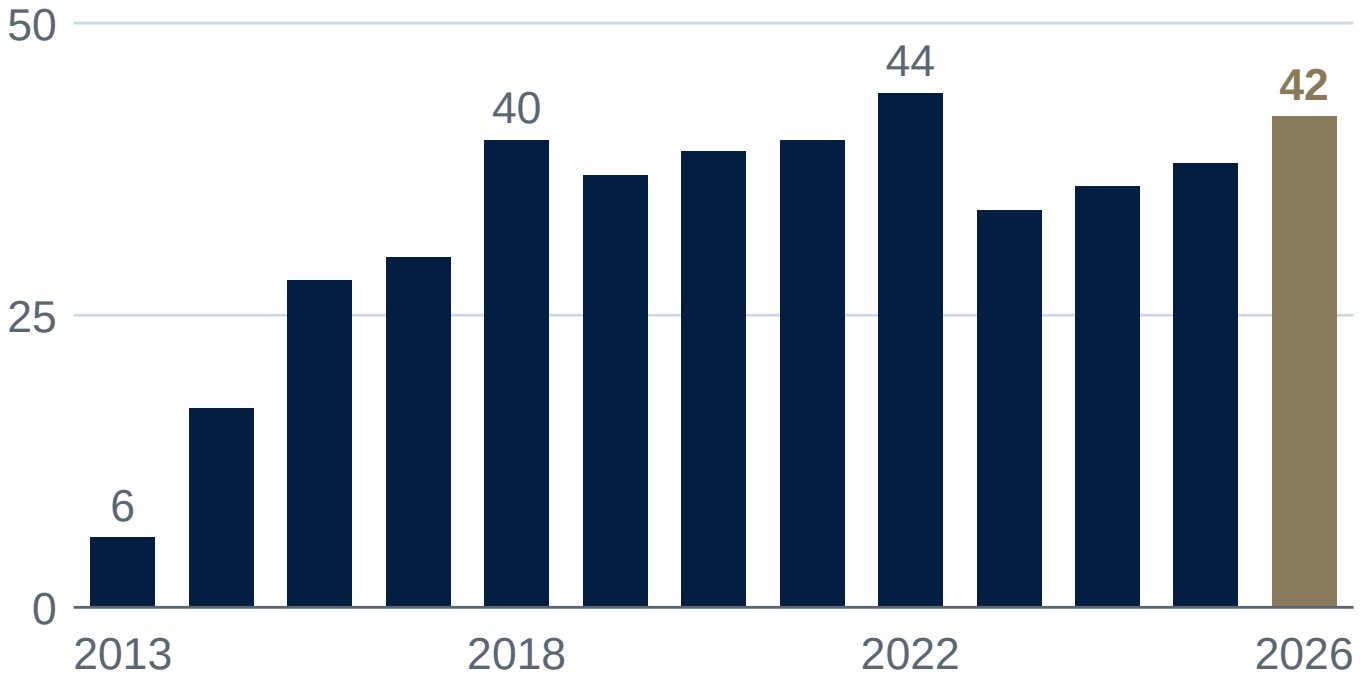
When a government names another government as the attacker, firms may lose their insurance. Should an attack by a state’s military hackers on private firms count as war?

> Who should bear the cost: the firm, its insurer, or its government?

Could it happen again?

Firms now rank cyber incidents as their top risk

PERCENT NAMING CYBER INCIDENTS A TOP RISK



Allianz Commercial, Allianz Risk Barometer, 2013 to 2026.

Cyber incidents have ranked first among business risks every year since 2022.

In 2026 artificial intelligence rose to second place.

> Concern rose as more of each firm's work came to run on shared software.

In 2025 one attack stopped Jaguar Land Rover

The carmaker shut its systems on September 2, 2025, and restarted production in stages from October 8.

More than 5,000 British firms were affected, and the government guaranteed a £1.5 billion loan.

> Eight years after NotPetya, one firm’s outage still spread through its suppliers.

JAGUAR LAND ROVER'S REVENUE IN THE QUARTER



JLR, results for the quarter to September 30, 2025, November 14, 2025.

Spend Maersk’s 2016 security budget

1. **Choose** one protection Maersk could have bought in 2016: dividing the network into sections, offline backups of the domain controllers, patching and replacing old Windows servers, or tying IT bonuses to security.
2. **Diagnose** which stage of the attack it would have stopped: getting in, spreading, or recovering, using one passage from Greenberg.
3. **Estimate** what it would have cost in a normal year against the $250 million to $300 million loss.
4. **Defend** your choice to the person next to you.

15 minutes · Individual, then pairs · Reading permitted

Which protection would have saved Maersk the most, and why had no one paid for it?

> Which stage did most of you choose, and what does that say about where firms are weakest?

Russia reached firms around the world through one country’s tax software. If one government controlled a network that most firms depend on, what could it do with that control?

> Next Wednesday Farrell and Newman ask who controls the networks global business depends on.

How do shocks travel through global business?

Firms cut spare stock and capacity to save money, so a delay at one step can stop a whole chain.

A shock travels along the links that make a firm efficient, such as shared ports in 2021 and a shared network in 2017.

Truckers, customers, and suppliers bear costs they did not choose, while whoever controls a bottleneck may gain.

Resilience costs money in normal years, so firms underbuy it until a shock shows its value.